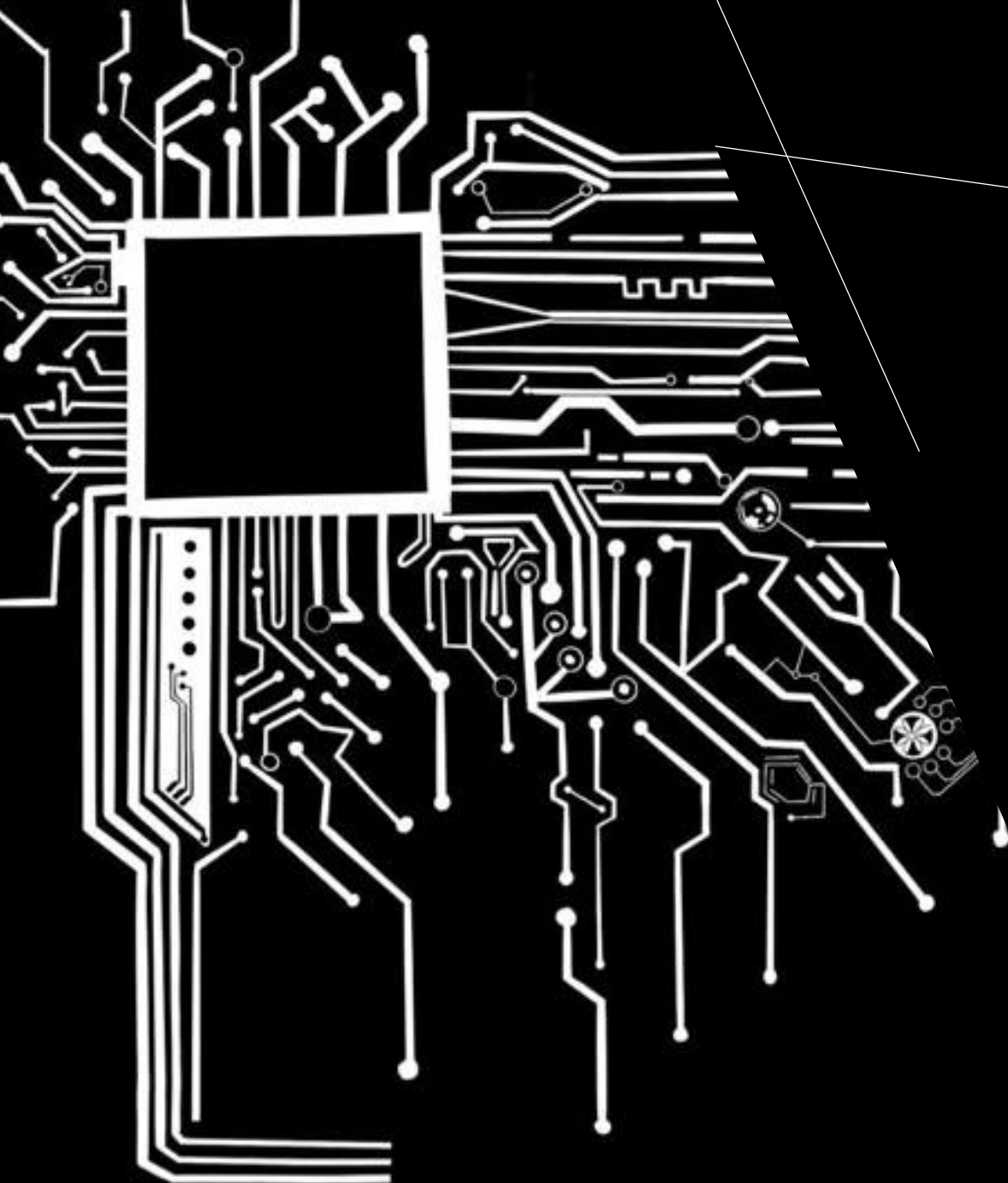




ICE 2343

DIGITAL LOGIC DESIGN

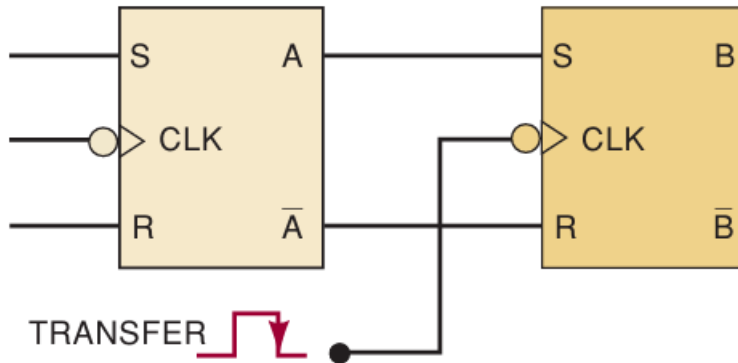
ABHIRUP SARKER
LECTURER
DEPARTMENT OF ROBOTICS AND MECHATRONICS
ENGINEERING
DAFFODILS INTERNATIONAL UNIVERSITY

Data Storage & Transfer

Most common use of flip-flop is for storing and transferring data.

How is the data stored?

The data is stored in binary form in group of flip-flops called **Register**. So, an n-bit register consists of n no. of flip-flop and capable of storing n bit data.



For example, two SR flip-flop in series used to transfer data. Flip-flop A stores data and for the negative edge trigger the data is transferred to flip-flop B.

Data Storage & Transfer

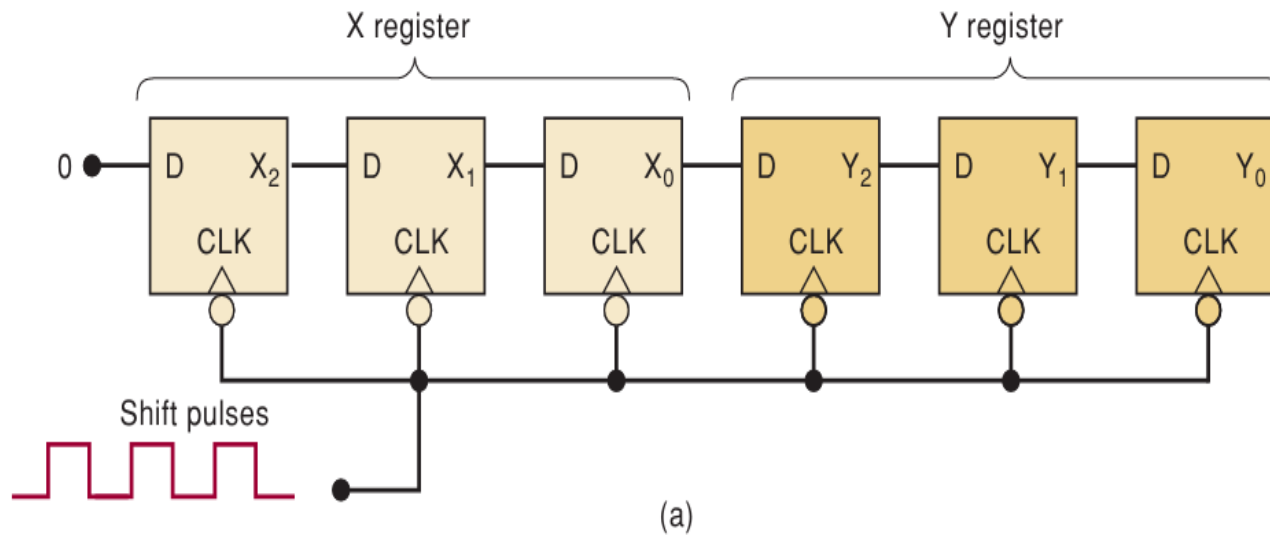
The data input to the flip-flops can be either serial or parallel input. Based on input and output the register is divided into four categories:

- I. Serial input serial output (SISO)
- II. Serial input parallel output (SIPO)
- III. Parallel input parallel output (PIPO)
- IV. Parallel input serial output (PISO)

Depending on application there are two types of register:

- 1. Shift register:** A shift register is a group of FFs arranged so that the binary numbers stored in the FFs are shifted from one FF to the next for every clock pulse. In SISO, SIPO and PISO data shift occurs hence they are shift registers.
- 2. Storage register:** A storage register is a group of FFs used to only store the data without shifting. Here the data is held simultaneously. PIPO is an example of storage register because the data is input parallelly and instantly available.

Shift Register

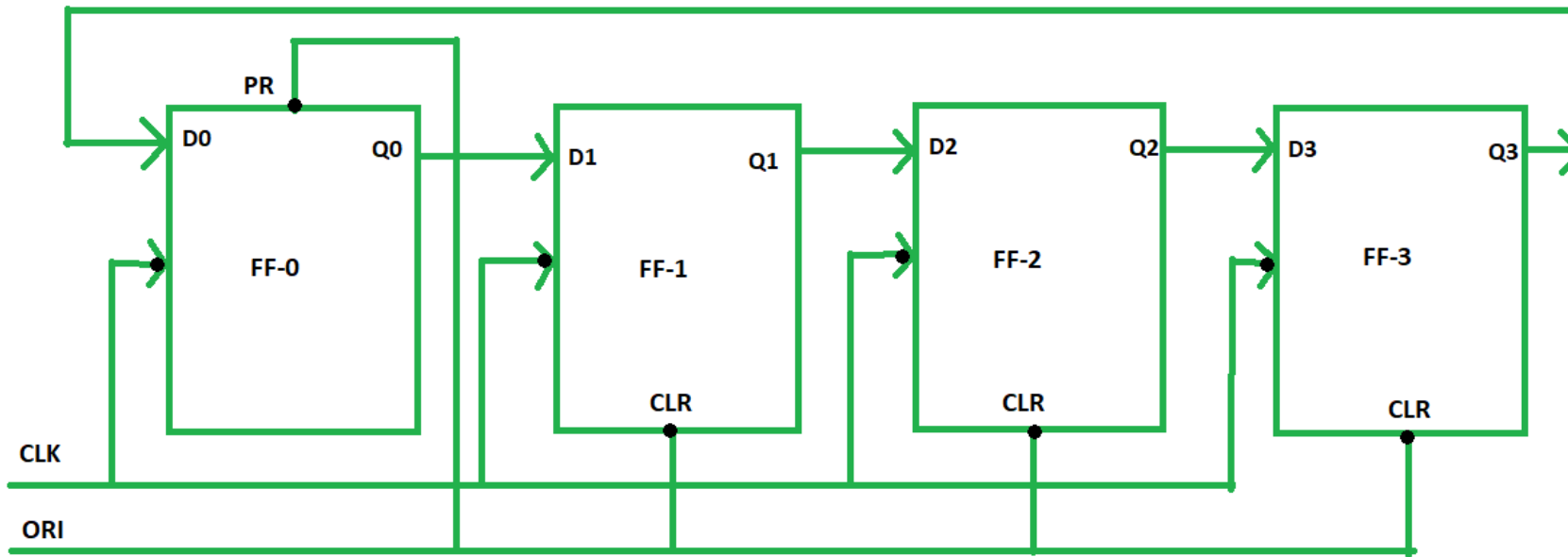


X_2	X_1	X_0	Y_2	Y_1	Y_0
1	0	1	0	0	0
0	1	0	1	0	0
0	0	1	0	1	0
0	0	0	1	0	1

- ❑ The three X FFs make X register and Y register consists of three Y FFs.
- ❑ D flip-flops have ability to retain data until explicitly changed, hence used in shift registers.
- ❑ X registers hold a certain data and with each clock pulse new data enters the FFs of X register and the previous data shifts to the right.
- ❑ This is why it is also called right shift register.

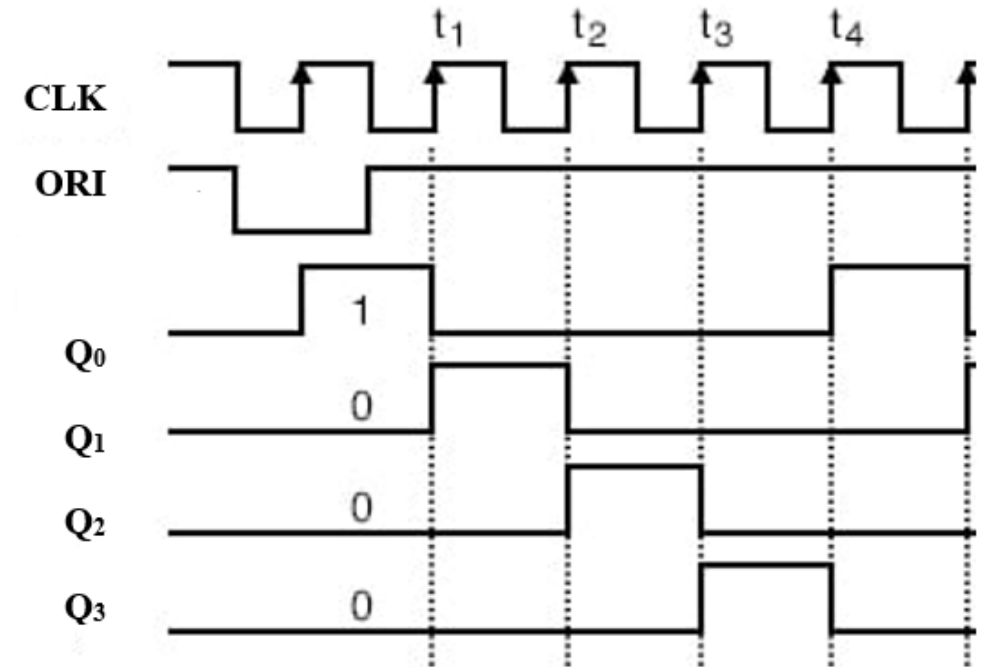
Ring Counter

- ❑ Ring counter is typically, a shift register where the output of last FF is input of first FF.
- ❑ So, it shifts 1 through series of FFs one at a time creating a loop. Hence, named ring counter.
- ❑ No. of states = no. of FFs used
- ❑ It has an active low overriding input (ORI) signal that enables preset (PR) and clear (CLR).
- ❑ When PR=0 output is set to 1 and when CLR=0 the output is set to 0.
- ❑ Ring counter is used where several sequential operation is controlled like resistance welding.



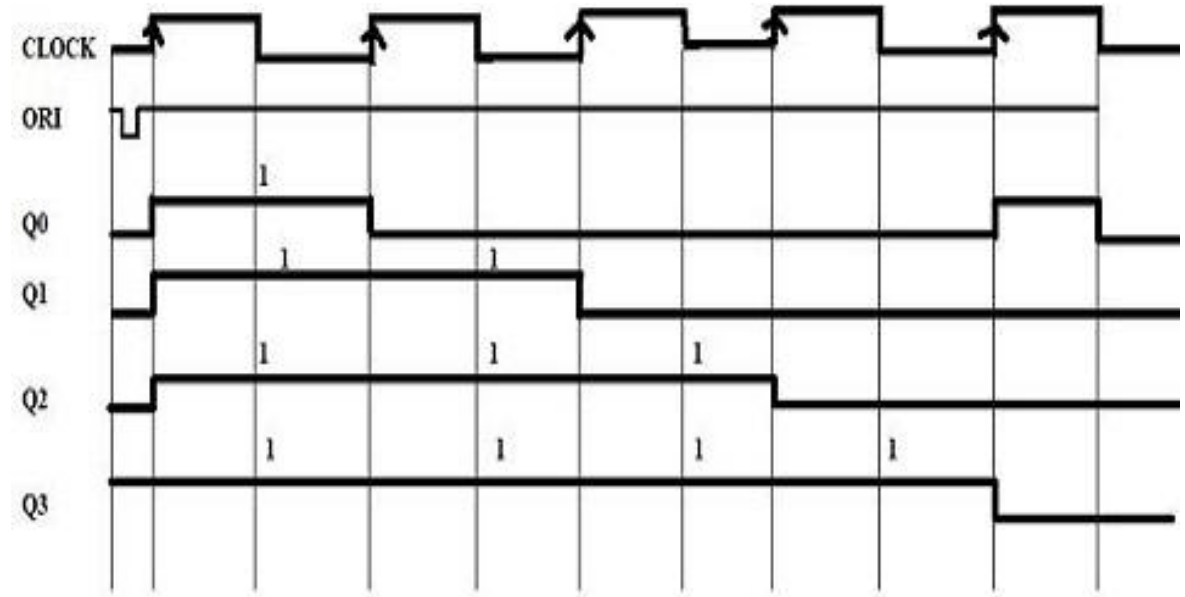
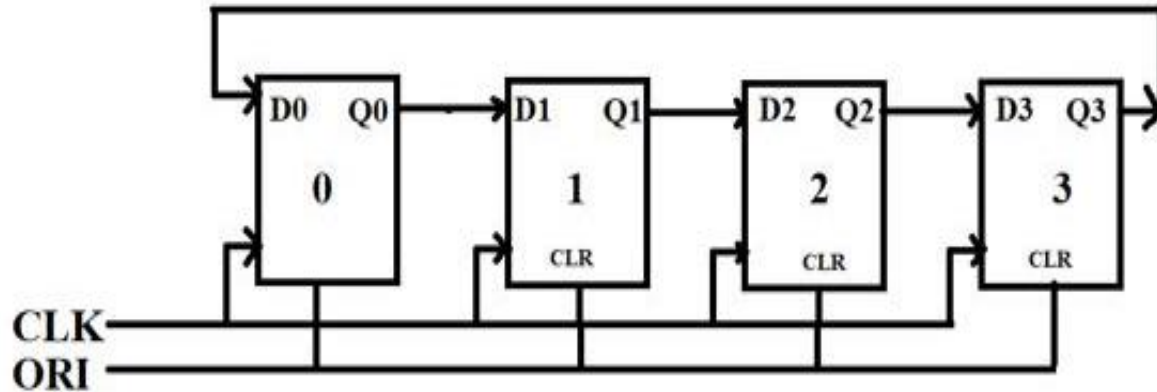
4-bit Ring Counter

ORI	CLK	Q0	Q1	Q2	Q3
Low	X	1	0	0	0
High	High	0	1	0	0
		0	0	1	0
		0	0	0	1



Johnson Counter

- ❑ Johnson counter operates by complemented output of the last FF feedback to first FF input.
- ❑ No. of states = $2 \times \text{no. of FFs}$
- ❑ It is also known as switch tail or twisted ring counter.
- ❑ It overcomes limitations of ring counter.

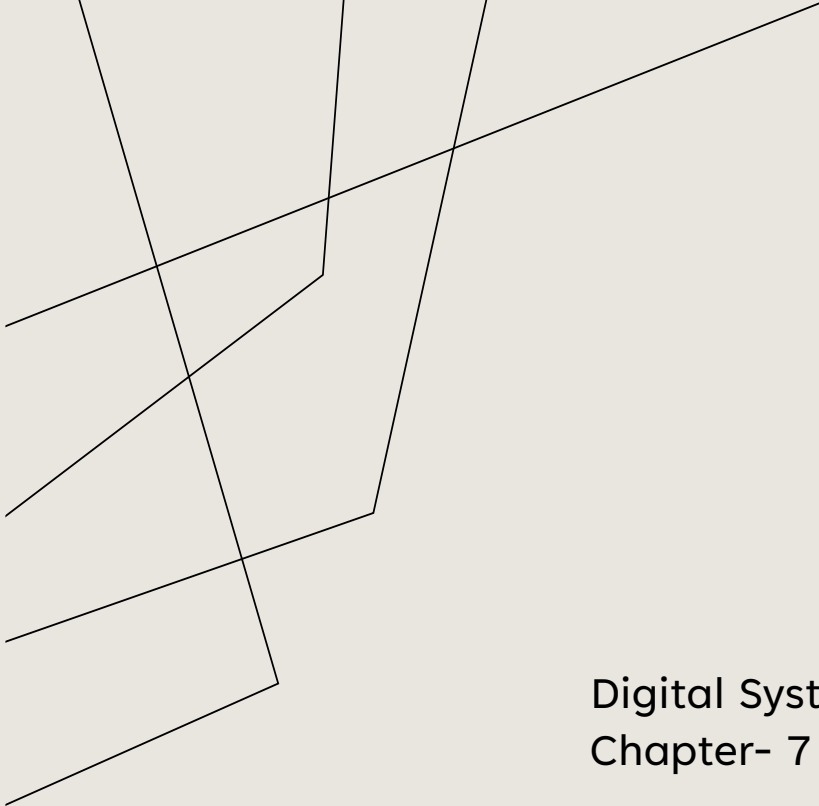


4-bit Johnson Counter

ORI	CLK	Q0	Q1	Q2	Q3
Low	x	0	0	0	0
High	High	1	0	0	0
		1	1	0	0
		1	1	1	0
		1	1	1	1
		0	1	1	1
		0	0	1	1
		0	0	0	1
		0	0	0	0

Ring Counter vs Johnson Counter

Ring Counter	Johnson Counter
Output of last flip-flop is input in first flip-flop	Complement of last flip-flop's output is input in first flip-flop
No. of states are equal to no. of flip-flops	No. of states are twice as no. of flip-flops
Only one bit can be high at a time	The output can be of any pattern
It counts binary sequence	It can not count binary sequence



Reference

Digital Systems Principles and Applications by Ronald J.Tocci
Chapter- 7

A series of white, thin, overlapping geometric lines on a black background, forming a complex, abstract shape on the left side of the slide.

THANK YOU